

# Japanese-language newspapers and OCR quality in Chronicling America

The missing Japanese text, our OCR of it, what other collections hold, and a sample audit of LoC's OCR in other languages

October 2026

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## Summary

- **LoC has no searchable Japanese text for these papers.** The Japanese pages of the WWII camp newspapers and of Denver's *Rocky Shimpō* and *Colorado Times* arrive from the Library of Congress with no text, or with garbled Latin characters. LoC's files show that the original OCR (ABBY 8) ran in Japanese on these pages and returned no words (116 of 124 sampled pages; the other 8 were run as English). About 9,330 pages across 23 titles have no text.
- **We read them ourselves.** NDLOC-Lite, the National Diet Library's open OCR engine, scored best of the engines we tested. Our job has read 11,018 of the 11,058 target pages (the targets also include pages whose LoC text is empty, short or garbled, which is why there are more than 9,330). The first 10,414 become searchable on usnewsmag.com with the index version now being built.
- **For most of these pages we found no Japanese text anywhere else, but the sites likeliest to hold the camp papers could not be searched yet.** We checked other collections that hold these papers:
  - *Rocky Shimpō*: Hoji Shinbun (Hoover Institution) has Japanese OCR for April 1943 to April 1944. Colorado Historic Newspapers has Japanese OCR for 1943 and part of 1944; where its run ends is not yet checked. LoC's run, and so ours, covers June 1944 to December 1945, so ours is new unless CHNC's run goes past June 1944.
  - *Colorado Times*: Colorado Historic Newspapers already has Japanese OCR, including 1945, so ours is a second reading, useful for comparison.
  - *Camp newspapers*: no Japanese OCR in any collection we could check. The sites most likely to have them (Densho, Utah Digital Newspapers, Wyoming Newspapers and others in the table) could not be searched automatically and are still to be checked by hand.
- **LoC's OCR in the other languages, from a 10% sample.** We sampled 2,371,337 of 23.7 million pages. Among English, German, Spanish and French, damage is highest in English pages from the 1840s to the 1880s, German pages before 1920, Spanish pages from the 1890s to the 1910s and French pages from the 1900s and 1910s, and lower from the 1920s on. In titles with several catalog languages, 46.8% of pages are in the first-listed language and 48.3% in another listed language or English.

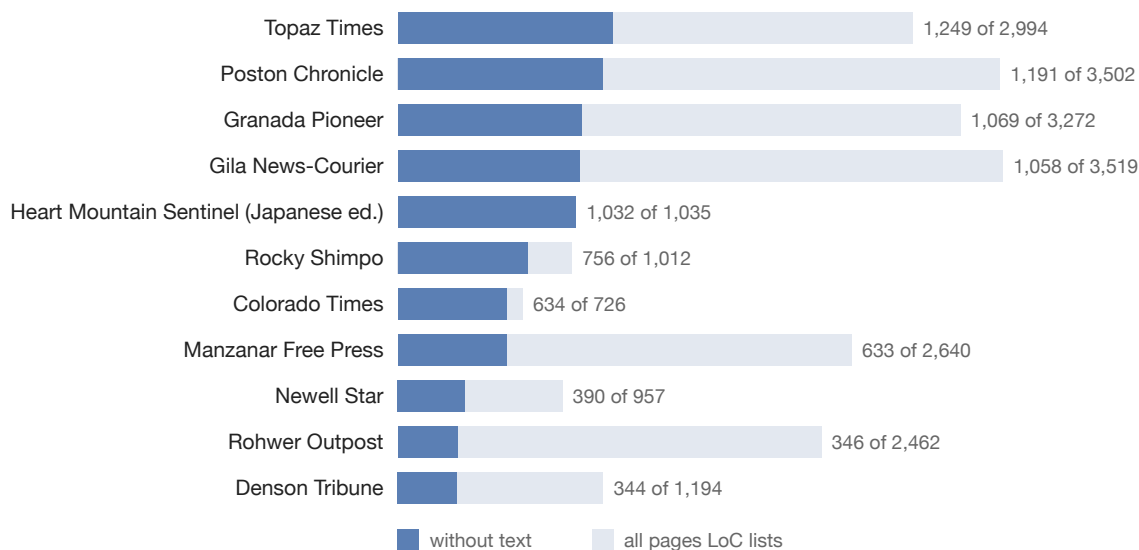
## The gap at LoC

Chronicling America lists 27 titles as Japanese. They are mostly the newspapers of the WWII incarceration camps (*Minidoka Irrigator*, *Manzanar Free Press*, the Japanese edition of the *Heart Mountain Sentinel*, *Poston Chronicle*, *Gila News-Courier*, *Topaz Times*, *Granada Pioneer*, *Tulean Dispatch*, *Rohwer Outpost*, *Newell Star* and others) plus two Denver papers, *Rocky Shimpō* and *Colorado Times*. The page images are

fine: LoC serves them at full scan resolution through IIIF (all but 14 pages, which have no image at LoC). The text is missing:

- **In the bulk OCR archives**, a Japanese page has an empty ALTO `ocr.xml` and no `ocr.txt`. In one batch, `dlc_ballston_ver01`, 2,433 of 5,791 pages are like that.
- **On loc.gov**, the page viewer shows “NO TEXT AVAILABLE FOR THIS PAGE”, and full-text searches across Chronicling America for 年, 日本 and 真珠湾 return no pages.
- **LoC’s files record what was run**. The empty ALTO keeps its processing settings. All 124 pages we sampled record `Engine:Abbyy8: 116` with `Lang:ja` and `Word Count:0`; the other 8 were run as English, which is where the garbled Latin text comes from. The Japanese OCR was run and read nothing.
- **The gap dates from 2014**. Four of the six batches with these papers were ingested in December 2014 and the other two in June 2026. All six were rebuilt in June 2026 and still have no Japanese text. NDNP’s technical guidelines have listed Japanese as a supported full-text language since at least the 2016–18 edition.

Comparing LoC’s per-title page counts, which include pages without text, with the pages that have text gives the size of the gap: about 9,330 Japanese pages across 23 titles. Every other non-English language we checked has its text (German, Spanish, French, Polish, Yiddish, Russian, Serbian, Czech, Hebrew).



Pages with no OCR text at LoC (dark) against all pages LoC lists for the title (light), for the 11 of 23 Japanese titles with the largest gaps. Per-title audit, 4 October 2026 (#135).

## What other collections hold

We searched other collections for Japanese OCR of the same papers. Where a site allowed it, the test was to find the title, search it for a common word (日本, “Japan”) and for exact lines from LoC’s pages, and compare the dates covered.

| Collection                                           | Holds                                                                                                                                                     | Japanese OCR                                                                    | What it means for our OCR                                                                                    |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Hoji Shinbun Digital Collection (Hoover Institution) | <i>Rocky Shimpō</i> 1943-04-12 to 1944-04-12 (101 issues, 404 pages); camp magazines and directories; not <i>Colorado Times</i> , not the camp newspapers | Yes, for what it holds                                                          | No overlap: LoC's <i>Rocky Shimpō</i> starts 1944-06-02                                                      |
| Colorado Historic Newspapers (CHNC)                  | <i>Rocky Shimpō</i> 1943–1944; <i>Colorado Times</i> 1918–1969                                                                                            | Yes (日本: 620 results in <i>Rocky Shimpō</i> , 72,008 in <i>Colorado Times</i> ) | <i>Colorado Times</i> 1945 is already OCR'd there; CHNC's last <i>Rocky Shimpō</i> date is not yet confirmed |
| National Diet Library Digital Collections            | One <i>Minidoka Irrigator</i> issue, English                                                                                                              | No                                                                              | None                                                                                                         |
| Internet Archive                                     | <i>Rohwer Outpost</i> , 115 items, 1942–44                                                                                                                | No (English OCR only)                                                           | None                                                                                                         |
| HathiTrust                                           | No records for these titles                                                                                                                               | n/a                                                                             | None                                                                                                         |
| Densho Digital Repository                            | Camp newspapers, all ten camps                                                                                                                            | Unknown (not searchable automatically)                                          | To check by hand                                                                                             |
| Utah Digital Newspapers                              | <i>Topaz Times</i> , about 470 issues                                                                                                                     | Unknown (not searchable automatically)                                          | To check by hand                                                                                             |
| Wyoming Newspapers                                   | <i>Heart Mountain Sentinel</i> , reportedly the Japanese edition too                                                                                      | Unknown (not searchable automatically)                                          | To check by hand                                                                                             |
| Arizona Memory Project / ASU, CDNC / Calisphere, CRL | Possibly <i>Poston</i> , <i>Gila</i> , <i>Tule Lake</i> , <i>Manzanar</i> (unconfirmed)                                                                   | Unknown (not searchable automatically)                                          | To check by hand                                                                                             |

### Rocky Shimpō

Hoji Shinbun

Japanese OCR, 404 pages

CHNC

? 1943 and part of 1944; dates not confirmed

LoC / ours

756 of 1,012 pages lack text at LoC; our OCR

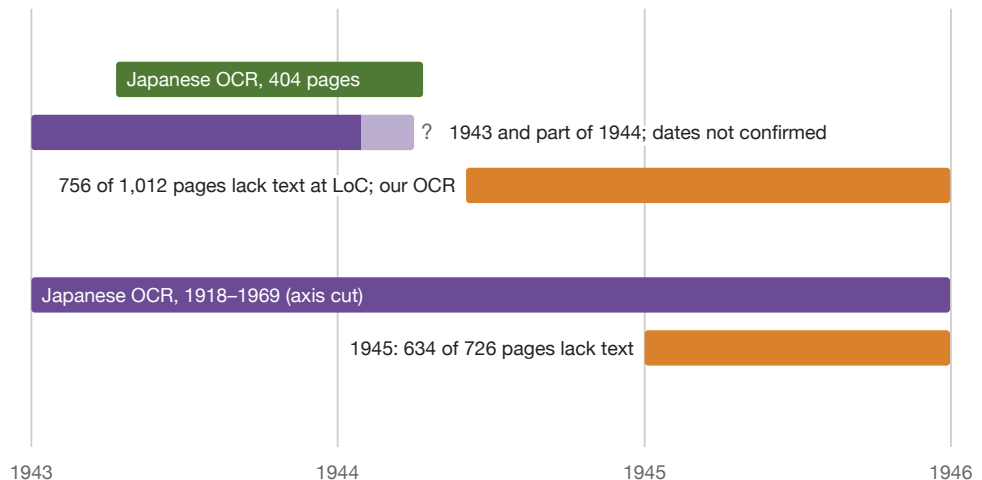
### Colorado Times

CHNC

Japanese OCR, 1918–1969 (axis cut)

LoC / ours

1945: 634 of 726 pages lack text



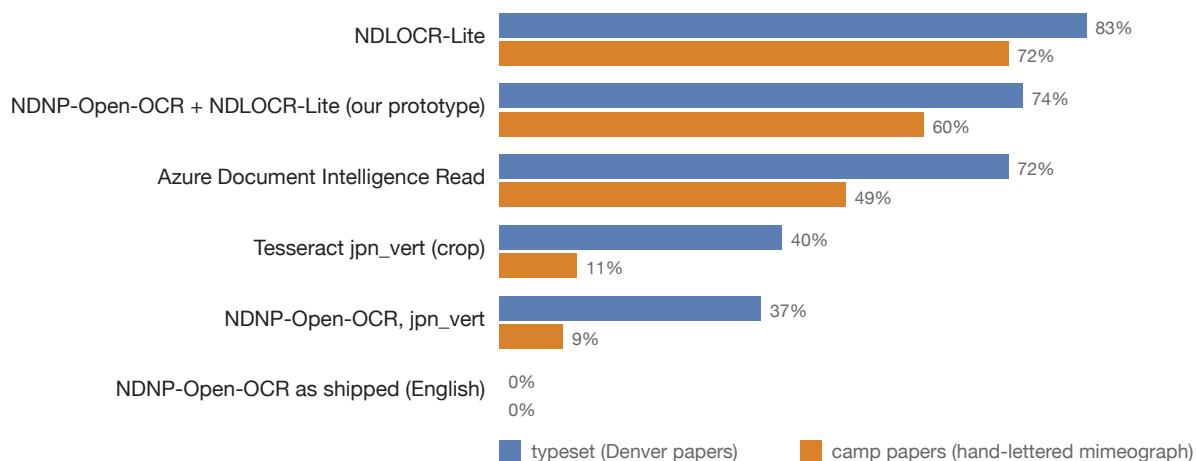
*Rocky Shimpō* and *Colorado Times*: which collection holds which dates, and whether it has Japanese text. CHNC's *Rocky Shimpō* dates come from search results by year and are not confirmed. CHNC's *Colorado Times* runs 1918–1969; the axis starts at 1943.

By paper:

- **Rocky Shimpō** is split in time. Hoji holds April 1943 to April 1944 with Japanese OCR, and CHNC holds 1943 and at least part of 1944; LoC holds June 1944 to December 1945 with no Japanese text. Hoji's run ends 1944-04-12 and LoC's starts 1944-06-02. Of the collections we checked, our OCR of LoC's 756 pages without text is the only Japanese text for those 19 months, unless CHNC's run extends past June 1944.
- **Colorado Times** is the one paper where our text duplicates existing work: CHNC has Japanese OCR across 1918–1969. An exact line from the 1945-08-14 issue (皆様お馴染の三光樓) found nothing on CHNC, probably because the two OCR readings differ, which makes the pair useful for comparing quality.
- **The camp newspapers** are not in Hoji, which instead holds camp literary magazines and directories (*Dotō* and *Gyōen* from Tule Lake, an Amache directory) that LoC doesn't have. We found no Japanese OCR of the camp newspapers anywhere we could search. The sites likeliest to have it could not be checked automatically.

## Our OCR

**Choosing an engine.** We cut 23 reference regions from 124 sampled Japanese pages of 17 titles (1942–45), split between the typeset Denver papers and the camp papers (mostly hand-lettered mimeograph), and scored each engine by bigram F1, the overlap of adjacent character pairs between the engine's text and the reference, which measures whether the words are there, as search needs. NDLOCR-Lite (CC BY 4.0, runs on CPU) scored highest overall, 77% against Azure's 61%, and beat Azure Document Intelligence on 19 of the 23 crops. We also tried LoC's reprocessing pipeline, NDNP-Open-OCR (commit `cbe0ca5`). It is set up for English only, so it read every page as English. With Japanese Tesseract models it reached 24%, and with NDLOCR-Lite swapped in as its engine (our local prototype, not proposed upstream) 67%. The pipeline runs on full pages, and scoring only the text inside each crop costs it some points against the crop-only scores (#128).



Bigram F1 (overlap of adjacent character pairs with the reference) on the 23 reference crops, typeset and camp papers. NDNP-Open-OCR rows ran on full pages and keep only text inside each crop, which lowers their scores somewhat. The prototype is ours, not LoC's.

**The job.** For each issue with target pages, the job downloads the full-resolution images from loc.gov, runs NDLOCR-Lite, and writes the text twice: as ALTO in the same units as LoC's empty files, so it could stand in for them, and as one row per page for our index. It paces itself to loc.gov's limits. Targets are the pages LoC ships without text plus pages whose LoC text is empty, short or garbled: 11,058 pages in 3,421 issues.



From a LoC page image to a search result.

### Where it stands (6 October 2026).

- **Read:** 11,018 of 11,058 target pages.
- **Not read:** the other 40 pages. 14 of them have no image at LoC. The job writes an issue only when every page in it is read, so pages that share an issue with them are held back too.
- **Search:** the index version being built now includes the 10,414 pages read before it started; the other 604 follow with the next release. A search in Japanese script searches only these pages, with old character forms matched to modern ones (戦 finds 戦).

## LoC's OCR in other languages

**How we measured.** We scored a fixed 10% sample of the pages in our published index (version `pages-v20261003-1`): 2,371,337 of 23.7 million pages, from all 2,989 batches, on 7 October 2026. The 2,426 sampled pages of titles that list Japanese were counted but not scored. For each page the audit does two things.

- **It finds the page's own language.** The candidates are the title's catalog languages plus English. The page's language is the one whose 100 most frequent words make up the largest share of the page's words. A page is *undetermined* when it has too few of those words or two languages come out too close. It is *mixed* when stretches of it fall clearly to two different languages.
- **It measures damage in that language.** Of all occurrences of the language's 20 most common words, the damage rate is the share that come out one OCR edit wrong, such as "tbe", "tlie" or "aud" for "the" and "and". Misreadings that are real words don't count. We call a page *damaged* above 0.1 and *badly damaged* above 0.25. Even good text scores above zero: the lowest median for any language and decade with 100 or more scored pages is 0.043 (English, 1930s).

The word lists come from wordfreq, except Yiddish (from Yiddish Wikipedia and Wikisource) and Hawaiian (from the Hawaiian Corpus Project), which wordfreq doesn't cover. Serbian pages in Cyrillic are scored letter for letter in Latin script. A 2% sample run the day before, whose pages are all in the 10% sample and which was scored before the Yiddish, Hawaiian and Cyrillic Serbian word lists were added, agreed to within a tenth of a percentage point for English and German overall, and within half a point for each decade of English from the 1860s to the 1950s.

**Page language against catalog language.** Chronicling America catalogs languages per title, so every page of a title gets the same language. On the sample:

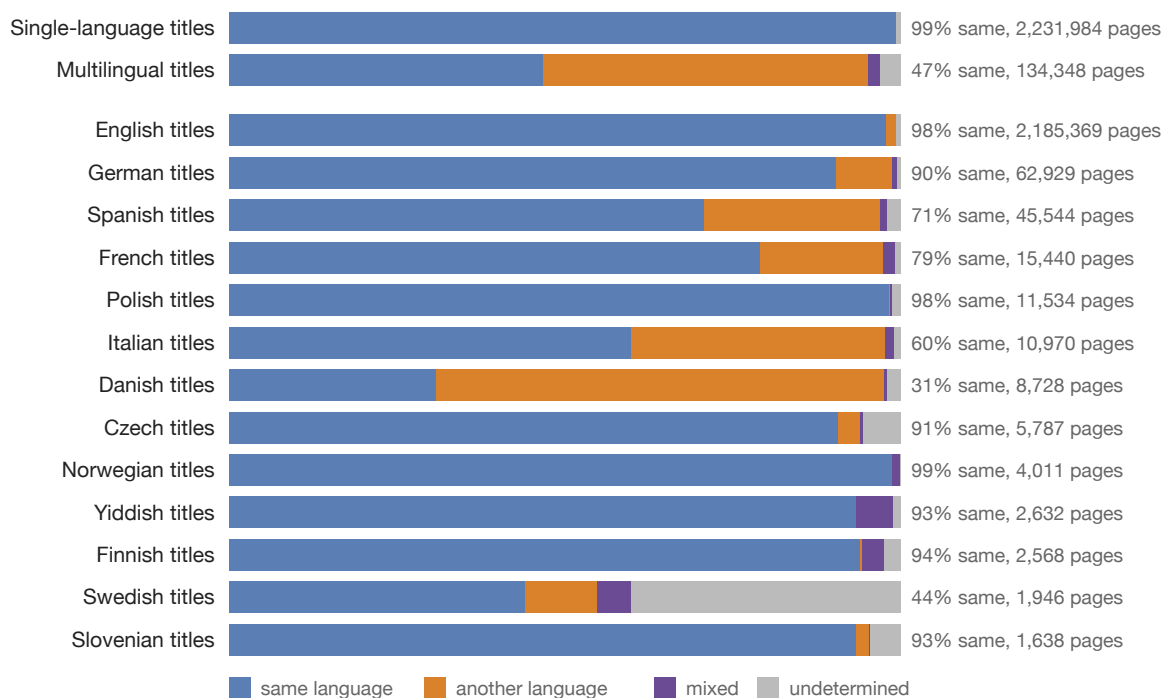
- **Single-language titles:** 99.3% of pages are in the title's language. Almost all the rest are undetermined (0.7%), and only 0.03% are in another language. For these titles English is the only other candidate (none for English-only titles), so the 99.3% mostly measures how many pages the audit could place at all.
- **Multilingual titles:** 46.8% of pages are in the first-listed language. Another 48.3% are in a different listed language or English.

Some of the largest groups of pages in a language other than the title's first:

- English pages in titles that list Spanish first: 11,919 sampled pages, 26% of those titles' pages.
- English pages in titles that list Danish first: 5,827, 67%.
- English pages in titles that list German first: 5,225, 8%.
- Spanish pages in titles that list Italian first: 3,893, 35%.
- English pages in titles that list French first: 2,814, 18%.
- Pages of titles that list English first and another language as well: 8,763 in Yiddish, 6,227 in German, 5,050 in Polish and 4,312 in Serbian.

The site's Hawaiian language filter returns four titles, the *Polynesian*, *Hilo Tribune*, *The Garden Island* and *The Maui News*, all English-language papers. Of the sampled pages in titles that list Hawaiian first, 98.7% are English and about 1% mix in Hawaiian. Choctaw, Dakota and Navajo titles also show as English or undetermined. Cherokee titles show as English (27 of 33 sampled pages) or as English mixed with Cherokee (6), because the audit recognizes the Cherokee syllabary by its script. With no word lists for these languages, the audit cannot see Choctaw, Dakota or Navajo text, so a page shown as English may also carry it.

Overall, 3.7% of sampled pages with text are not clearly in their title's first language: 2.8% are in another language, 0.8% are undetermined and 0.16% are mixed. If the sample holds, that is roughly 880,000 pages, about 690,000 of them in another language or mixed.



Detected page language against the title’s first catalog language, for sampled pages. Below the gap, titles grouped by first catalog language, for languages with at least 1,500 sampled pages.

**Damage by language and decade.** Detected languages with 4,000 or more scored pages:

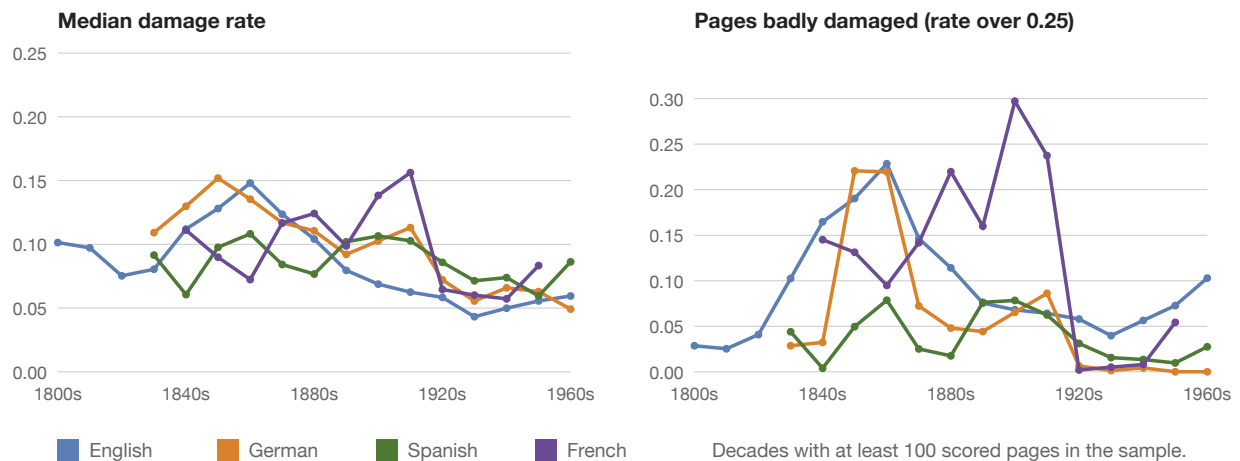
| Detected language | Scored pages | Median damage | Damaged (over 0.1) | Badly damaged (over 0.25) |
|-------------------|--------------|---------------|--------------------|---------------------------|
| English           | 2,166,233    | 0.071         | 35%                | 8.1%                      |
| German            | 63,086       | 0.105         | 53%                | 7.1%                      |
| Spanish           | 38,696       | 0.084         | 36%                | 4.1%                      |
| Polish            | 16,379       | 0.178         | 88%                | 16.8%                     |
| French            | 13,109       | 0.075         | 36%                | 10.3%                     |
| Yiddish           | 11,219       | 0.160         | 91%                | 13.8%                     |
| Italian           | 8,376        | 0.114         | 60%                | 4.1%                      |
| Czech             | 5,315        | 0.212         | 96%                | 39.2%                     |
| Norwegian         | 4,929        | 0.084         | 31%                | 3.0%                      |
| Serbian           | 4,312        | 0.257         | 99%                | 52.7%                     |

Polish, Czech and Serbian score high in every decade, as do Finnish, Lithuanian, Slovak and Russian (overall medians 0.194, 0.173, 0.192 and 0.240). Our unchecked explanation: in these heavily inflected languages many real word forms one edit from a common word are outside the 5,000 most frequent words the audit treats as real, so they count as damage. Yiddish may score high for a related reason (also unchecked): its word list is mostly modern spelling, and American Yiddish papers before the 1930s often

spelled words the older, German-influenced way (די for י). The decade figures do not show a drop after the 1920s, though. Compare these languages only with themselves over time.

By decade:

- **English:** damage peaks in the 1860s (median 0.148; 70% of pages damaged, 23% badly) and falls steadily to 0.043 in the 1930s.
- **German:** damage stays at 0.09 to 0.15 from the 1850s through the 1910s, with 46% to 78% of pages damaged. It then drops to 0.072 in the 1920s and 0.056 in the 1930s. That timing fits German-language papers moving from Fraktur to roman type around the First World War, but we have not checked the typefaces.
- **French:** damage is high in the 1900s and 1910s (0.138 and 0.156, with 30% and 24% badly damaged).
- **Yiddish:** damage stays between 0.155 and 0.176 from the 1910s through the 1940s, then is 0.131 in the 1950s.



Median damage rate (left) and share of pages badly damaged (right) by decade, for English, German, Spanish and French.

**Most damaged titles and batches.** These are the titles with the highest median damage, among titles with at least 50 scored pages:

| Title                                         | Place                           | Language | Scored pages | Median damage |
|-----------------------------------------------|---------------------------------|----------|--------------|---------------|
| <i>Nebraska Staats-Zeitung</i>                | Nebraska City and Lincoln, Neb. | German   | 292          | 0.62          |
| <i>Corpus Christi Caller and Daily Herald</i> | Corpus Christi, Tex.            | English  | 714          | 0.55          |
| <i>The Daily Wabash Express</i>               | Terre Haute, Ind.               | English  | 191          | 0.54          |
| <i>Council Bluffs Bugle</i>                   | Council Bluffs, Iowa            | English  | 140          | 0.51          |
| <i>The Spirit of the Age</i>                  | Woodstock, Vt.                  | English  | 61           | 0.51          |
| <i>Weekly Council Bluffs Bugle</i>            | Council Bluffs, Iowa            | English  | 58           | 0.49          |
| <i>True American</i>                          | New Orleans, La.                | English  | 146          | 0.48          |

| Title                                                  | Place                | Language | Scored pages | Median damage |
|--------------------------------------------------------|----------------------|----------|--------------|---------------|
| <i>The Texas Republican</i>                            | Marshall, Tex.       | English  | 384          | 0.48          |
| <i>Brownlow's Knoxville Whig, and Rebel Ventilator</i> | Knoxville, Tenn.     | English  | 76           | 0.47          |
| <i>The Corpus Christi Caller</i>                       | Corpus Christi, Tex. | English  | 799          | 0.46          |

The five batches with the highest median damage are `nn_kant_ver01`, `txdn_japan_ver01`, `nn_bentham_ver01`, `nn_carson_ver02` and `kyu_frenchie_ver02`. Each has a median damage of 0.46 to 0.52, and 88% to 99.8% of their scored pages are badly damaged. All of their scored pages are English; `txdn_japan_ver01` is LoC's batch name, not a language. All five were among the ten highest in the 2% sample.

The titles with the most undetermined pages are:

- *Omnibus* (German): 94% of 111 sampled pages with text
- *Skaffaren och Minnesota Stats Tidning* (Swedish): 88%
- *America* (Cleveland; English and Romanian): 83%
- *Minnesota Stats Tidning* (English and Swedish): 81%
- *Skaffaren* (Swedish): 81%

Across all titles that list Swedish or Hungarian first, 40% and 44% of pages with text are undetermined. We have not checked why. Damaged text, short pages, or text split between the listed languages can each leave a page undetermined.

### What it suggests for usnewsmap.com.

- **Tag page language only where it differs from the title's first language.** That is about 3% of pages, those in another language or mixed. As far as the audit can tell, the title's language stays right for 99% of pages in single-language titles. Languages the audit has no word list for keep their title language.
- **Pilot a German re-OCR with a Fraktur model.** German pages from the 1850s to the 1910s are an estimated 580,000 pages at full scale. Score a sample the same way before and after.
- **Re-OCR the worst batches whole.** The five batches above are the starting point.

## Caveats

- **The accuracy figures are agreement with a machine reference.** The reference transcriptions were machine-made and one was checked against its scan. The ranking between engines is better established than any one percentage. A Japanese reader should review a sample of the crops and of NDLOCR-Lite's output.
- **Camp papers are harder for every engine.** They are mostly hand-lettered mimeograph, and NDLOCR-Lite scores 72% on them against 83% on typeset pages.
- **Other collections were checked only by searching their public websites,** and the coverage table reflects what those sites showed on 4 and 5 October 2026. Absence of a search hit is not proof of

absence.

- **The OCR audit is a 10% sample.** Per-title figures need at least 50 sampled pages: scored pages for damage, pages with text for the undetermined list. The damage rule catches only one-edit misreadings of common words, so it undercounts heavier errors, and even good text scores above zero. The audit only considers a title's catalog languages plus English, so it cannot find pages in a language the catalog leaves out. Choctaw, Dakota, Navajo and Cherokee have no word lists, so text in them is never scored: their titles' pages come out as English (scored against English) or undetermined, or, for Cherokee printed in its syllabary, as mixed. The Yiddish and Hawaiian lists are our own, from open corpora; how well they suit 19th- and early 20th-century print is unchecked.

## Open questions

1. **CHNC's last *Rocky Shimpō* issue.** If it is before June 1944, our *Rocky Shimpō* text is new throughout.
2. **Densho, Utah Digital Newspapers, Wyoming Newspapers and the other sites in the table,** checked by hand: search 日本 within each camp title.
3. **A quality comparison with CHNC** on the same *Colorado Times* 1945 pages.
4. **A Japanese reader's review** of the reference crops and a sample of our OCR.
5. **Typefaces of the damaged German pages:** how many of the German pages from the 1850s to the 1910s are Fraktur, and how much a Fraktur model recovers.

## Sources

- Progress on 6 October 2026: the usnewsmap.com status API (<https://api.usnewsmap.com/v1/status>, section ocr\_ja: 11,018 of 11,058 pages read). The 10,414 pages in the index being built are those the job had written when the build read the OCR output, at 6:21 AM ET on 6 October, as recorded in [#187](#).
- usnewsmap.com issues [#128](#) (OCR), [#135](#) (the gap and other collections) and the technical note *Japanese pages: OCR of the pages LoC ships without text* (<docs/notes/2026-10-05-japanese-ocr.md>).
- Library of Congress, [Japanese-American Internment Camp Newspapers](#); NDNP Technical Guidelines [2025–27](#) and [2016–18](#); [NDNP-Open-OCR](#).
- [Hoji Shinbun Digital Collection](#), Hoover Institution; [Colorado Historic Newspapers](#).
- [NDLOCR-Lite](#), National Diet Library.
- OCR quality audit: `ja-ocr/quality.py` (metric v2) in the usnewsmap.com repository, run on index version `pages-v20261003-1` at 10% on 7 October 2026 (code at commit `480f8e4`) and at 2% on 6 October 2026 (before the Yiddish, Hawaiian and Cyrillic Serbian word lists were added). Its table rows are in `data/ocr-quality-v2-10pct.jsonl` and `data/ocr-quality-v2-2pct.jsonl` next to this report. Word lists: [wordfreq](#); for Yiddish, Yiddish Wikipedia and Wikisource (CC BY-SA 4.0); for Hawaiian, the [Hawaiian Corpus Project](#) (CC0).